

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech. Branch: Civil Engineering Semester: VIIth Semester

Subject Code & Name: BTCVE705D Rock Mechanics

Max Marks: 60

Date: 11-01-24

Duration: 3.00 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

Marks

Q. 1 Solve Any Two of the following.

- | | |
|--|----------|
| A) Write short notes on: - | 6 |
| a. Lineaments | |
| b. Discontinuities in Rocks | |
| c. History of rock mechanics | |
| B) What are its primary objectives and scope in the field of rock mechanics? | 6 |
| C) Explore the factors that affect the strength of rocks. | 6 |

Q.2 Solve Any Two of the following.

- | | |
|--|----------|
| A) Write a short note on: - | 6 |
| a. Primary purpose of rock testing | |
| b. Uniaxial compressive strength of rock | |
| c. Tri-Axial compressive test in rock testing | |
| B) Discuss purpose and significance of swelling and slake durability tests. | 6 |
| C) Explain the importance of tensile Strength of rocks and the methods used for its determination in the laboratory. | 6 |

Q. 3 Solve Any Two of the following.

- | | |
|--|----------|
| A) Write a short note on: - | 6 |
| a. Concept of rock mass | |
| b. NGI Classification System | |
| c. Rock Quality Designation (RQD) | |
| B) Discuss the factors influencing the behavior of rock masses at slopes. | 6 |
| C) Explain the Geological Strength Index (GSI) and its significance in rock mass classification. | 6 |

Q.4 Solve Any Two of the following.

- A)** Write a short note on: - **6**
 - a. Rock falls
 - b. Slope monitoring techniques
 - c. Rock slope prevention techniques
- B)** Explain the three main modes of failure in rock slopes with neat labeled diagrams. **6**
- C)** Describe the methods commonly used for the analysis of rock slope stability. **6**

Q. 5 Solve Any Two of the following.

- A)** Write a short note on: - **6**
 - a. Rock Reinforcement
 - b. Rock bolting
 - c. Shotcrete
- B)** Discuss the different types of grouting techniques and their applications in enhancing rock mass properties. **6**
- C)** Explain the key differences between the Mohr-Coulomb and Hoek and Brown criteria. **6**

***** End *****